

CUSTOM THREAT DETECTION ENGINEERING

Wazuh SIEM | Detection Engineering Case Study

PROJECT FOCUS

Closing detection gaps across web, Linux persistence, and privilege escalation activity.

ROLE

SOC trainee performing rule engineering, MITRE ATT&CK; mapping, and alert validation.

THE CHALLENGE

Three adversary-simulation events were present in the logs but lacked the required high-priority detection coverage. The task was to analyze each event, engineer precise custom Wazuh rules in the assigned ID range, map the detections to MITRE ATT&CK;, and prove that each rule generated the expected alert using **wazuh-logtest**.

MY APPROACH

01	02	03	04
Analyze Reviewed raw security events and identified the behavior requiring detection.	Engineer Built targeted XML logic using parent rules, PCRE2 matching, URL matching, and decoded fields.	Map Mapped each detection to MITRE ATT&CK; and assigned the required severity.	Validate Tested each event with wazuh-logtest and confirmed Phase 3 matches and alert generation.

DETECTION COVERAGE

Scenario	Rule	Level	MITRE	Security Goal
SQL Injection	100200	12	T1190	Detect URL-based SQL injection against a public-facing application.
Cron Persistence	100201	10	T1053.003	Detect non-root modification of scheduled cron tasks.
Sudo / Find Abuse	100202	12	T1548.003	Detect sudo execution of /usr/bin/find associated with privilege escalation.

DETECTION WORKFLOW



TECHNICAL EVIDENCE

Three custom detections - three successful Wazuh Logtest validations

DETECTION 01 - SQL INJECTION		Rule 100200 Level 12 MITRE T1190
<pre><rule id="100200" level="12"> <if_sid>31100</if_sid> <url type="pcre2">(?!admin%27%20R%201=1--</url> <description>SQL Injection attempt detected in URL</description> <mitre><id>T1190</id></mitre> </rule></pre>		Validation Phase 3 matched rule 100200 at Level 12 and mapped it to Initial Access / Exploit Public-Facing Application. PASS - Alert successfully generated
DETECTION 02 - CRON PERSISTENCE		Rule 100201 Level 10 MITRE T1053.003
<pre><rule id="100201" level="10"> <if_sid>2834</if_sid> <regex type="pcre2">\((?!root\))\s+\s+BEGIN EDIT</regex> <description>Non-root user modified crontab scheduled tasks</description> <mitre><id>T1053.003</id></mitre> </rule></pre>		Validation Phase 3 matched rule 100201 at Level 10 and mapped it to Cron under Execution, Persistence, and Privilege Escalation. PASS - Alert successfully generated
DETECTION 03 - SUDO FIND ABUSE		Rule 100202 Level 12 MITRE T1548.003
<pre><rule id="100202" level="12"> <if_sid>5403</if_sid> <field name="command" type="pcre2">^/usr/bin/find(\s \$)</field> <description>Potential privilege escalation using sudo and /usr/bin/find</description> <mitre><id>T1548.003</id></mitre> </rule></pre>		Validation Phase 3 matched rule 100202 at Level 12 and mapped it to Sudo and Sudo Caching under Privilege Escalation and Defense Evasion. PASS - Alert successfully generated

SECURITY VALUE

This exercise demonstrated the full detection-engineering lifecycle: interpreting raw telemetry, writing focused detection logic, mapping behavior to MITRE ATT&CK, assigning meaningful severity, and validating that the SIEM produces actionable alerts. The result was improved visibility across three different attack behaviors without relying solely on default detections.

SKILLS DEMONSTRATED	Wazuh SIEM • Detection Engineering • Log Analysis • XML Rule Development • PCRE2 • MITRE ATT&CK; • Linux Security • Web Attack Detection • Alert Validation
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Portfolio case study prepared from validated SOC residency assessment evidence. Personal trainee contact information is intentionally omitted from this public-facing version.